Regression:

t_i ith target value
x_i ith input value
$y(x)$ predicted value for x
$\boldsymbol{w}$= $(\mathrm{w}_0, \mathrm{w}_1, \dots, w_M)^T$vector of model parameters
$\boldsymbol{\phi}(x) = \left(\phi_0(x), \phi_1(x), \dots \phi_M(x)\right)^T$ vector of basis functions
$y(x) = \boldsymbol{w}^{\mathrm{T}}\boldsymbol{\phi}(x) = w_0\phi_0 + \cdots + w_M\phi_M$

Classification:

X_i ith feature
w_i ith weight
b model bias
$y(X_1, X_2) = w_1X_1 + w_2X_2 + b$ model prediction for point (X_1, X_2)

KKT conditions:

$\Sigma\, \alpha_n t_n = 0$ where α_n is the nth Lagrange multiplier
$\alpha_n(1 - y_n t_n) = 0$

Margin = $1/\left|\left|\boldsymbol{w}\right|\right|$

Clustering:
K-means kernels:

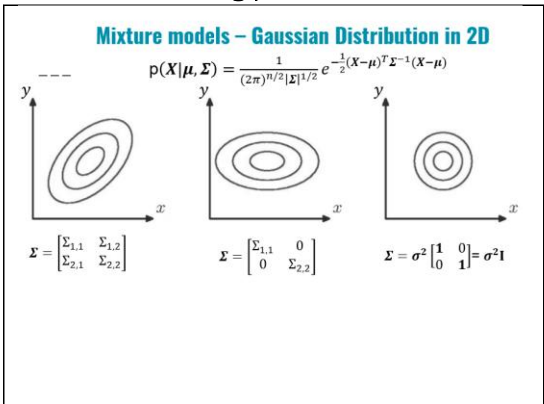
- Polynomial:

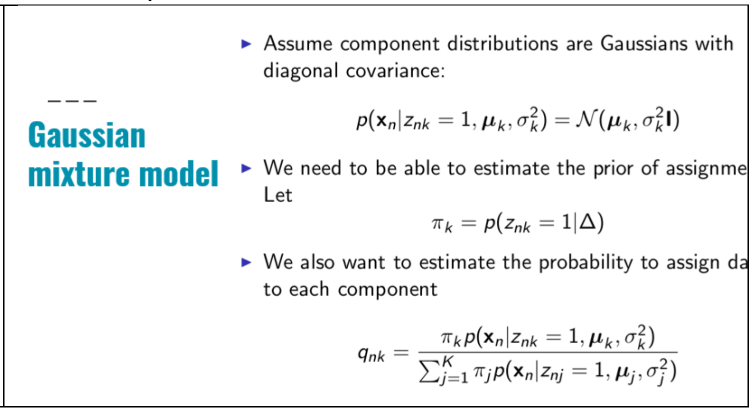
$$\kappa(x, y) = (x^{\mathsf{T}}y)^p$$

- Gaussian:

$$\kappa(x, y) = \exp(-\lambda|x - y|^2)$$

x, y: feature representations of data points
p: degree of the polynomial kernel
λ: Positive scaling parameter that controls the shape of the Gaussian kernel.





Mixture model optimisation – the Expectation-Maximization (EM) algorithm

–

- Following optimisation algorithm:
  1. Guess $\boldsymbol{\mu}_k, \sigma_k^2, \pi_k$
  2. **(E)**xpectation-step: Compute q_{nk}
  3. **(M)**aximization-step: Update $\boldsymbol{\mu}_k, \sigma_k^2, \pi_k$
  4. Return to 2 unless parameters are unchanged.
- Guaranteed to converge to a local maximum of the lower bound.

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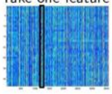
$$\widehat{N}_k = \sum_{i=1}^{N} q_{ik} \quad \hat{\mu}_k = \frac{1}{\widehat{N}_k}\sum_{i=1}^{N} q_{ik} x_i \quad \widehat{\boldsymbol{\Sigma}}_k = \frac{1}{\widehat{N}_k}\sum_{i=1}^{N} q_{ik}(x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

Projection:

Finding a subset – example

- Take one feature – N values.



- Some values from objects in class 1, some from class 0.
- Split them based on class and compute μ and σ^2 for each class.
- Compute s for each feature:

$$s = \frac{|\mu_1 - \mu_0|}{\sigma_0^2 + \sigma_1^2}$$

- Keep features with high s.

Principal Components Analysis

- Principal Components Analysis (PCA) is a method for choosing $\mathbf{W}$.
- It finds the columns of $\mathbf{W}$ one at a time (define the mth column as $\mathbf{w}_m$).
  - Each $D \times 1$ column defines one new dimension.
- Consider one of the new dimensions (columns of $\mathbf{Z}$):

$$\mathbf{z}_m = \mathbf{X}\mathbf{w}_m$$

- PCA chooses $\mathbf{w}_m$ to maximise the variance of $\mathbf{z}_m$

$$\frac{1}{N}\sum_{n=1}^{N}(z_{mn} - \mu_m)^2, \quad \mu_m = \frac{1}{N}\sum_{n=1}^{N} z_{mn}$$